

The totally symmetric $\zeta(7)$ cellular family:
a verified recurrence, two period ladders, the value of P_3 ,
and a Dwork-congruence reduction of the central-binomial law

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Abstract

We report the results of an intensive computational–theoretical campaign on Brown–Zudilin cellular integrals at weights five and seven. For the totally symmetric $\zeta(7)$ family on $\mathcal{M}_{0,10}$ we reconstruct an exact candidate recurrence for the leading coefficients q_n (order 4, degree 19, from 105 computed terms; verified against all 74 exact values), whose palindromic characteristic polynomial $\lambda^4 - 6340\lambda^3 + 67974\lambda^2 - 6340\lambda + 1$ factors over $\mathbb{Q}(\sqrt{3})$ into two reciprocal quadratic sectors. The family splits into two period ladders: conditionally, the weight-5 descent $(q_n, s_n, \hat{P}_n)$ rides the recurrence, while the primitive (P_n, I'_n) does not. Using the filtration identity $I_n = I'_n + \zeta(2) I''_n$ and the resulting conditional values $s_3, \hat{P}_3$, we determine, conditional also on the denominator grid, $P_3 = 23478462179525/(2^5 3^7)$ by a denominator-snap argument that requires *no* integral evaluation, verified independently (99.75% series climb; rigorous tail bound). Under those hypotheses, $\text{den}(P_3) \mid d_3^7$: the decisive 3-adic cell shows no denominator anomaly. Conditional sector measurements place both projections on the sector-B rate 0.0939; with the stated denominator cost, the standard criterion therefore cannot prove “one of $\zeta(5), \zeta(7)$ is irrational”. The $\zeta(2)$ -elimination costs a factor ≈ 594 in decay, the archimedean mirror of the Betti lattice cost. On the weight-5 arithmetic side, the primes $p \leq n$ of the central-binomial divisibility $\binom{2n}{n} \mid 6A_n$ (Phase 2) are reduced to a single new congruence: a vector-valued *Dwork descent* $p^5(p_n/q_n) \equiv p_a/q_a \pmod{p^{2-\kappa}}$, $a = \lfloor n/p \rfloor$, whose constants are the $n = 1$ ladder ratios $29/28 = p_1/q_1$, $101/84 = \hat{p}_1/q_1$; subject also to the required integer-lattice normalization at exceptional primes, the general inequality $\text{ord}_p(p_n) \geq \kappa - 5L$ is verified with zero failures. Every claim below is labeled proven, verified, or conjectural.

1 Setting and summary of status

Brown–Zudilin [1] construct cellular integrals I_n on $\overline{\mathcal{M}}_{0,N}$; at $N = 10$, the totally symmetric convergent cell $\sigma = (10, 2, 4, 1, 6, 3, 8, 5, 9, 7)$ yields a rank-4 motive with weight-graded pieces $\mathbb{Q}(0), \mathbb{Q}(-2), \mathbb{Q}(-5), \mathbb{Q}(-7)$ and linear forms

$$I'_n = \frac{75}{4} q_n \zeta(7) - 3 s_n \zeta(5) - P_n, \quad I''_n = -9 q_n \zeta(5) + 2 s_n \zeta(3) - \hat{P}_n, \quad I_n = I'_n + \zeta(2) I''_n,$$

with printed anchors $q_n = 1, 61, 52921$; $s_n = 0, 300, 261153$; $P_n = 0, 220, 6021219/32$; $\hat{P}_n = 0, 152, 535857/4$ for $n = 0, 1, 2$ [1]. The leading coefficients q_n are computable for all n by the McCarthy–Osburn–Straub diagonal construction [3], verified against the anchors and extended here to $n \leq 104$.

Status ledger. Throughout, *proven* = complete proof; *verified* = exact finite computation, no proof for all n ; *conditional* = exact consequence of named hypotheses.

Result	Status	Where
(W) : $w_n \equiv \tilde{w}_n \equiv 0 \pmod{p}$, $n < p \leq 2n$	proven (all n)	[4]
Phase 1 of $\binom{2n}{n} \mid 6A_n$	proven (all n)	[4]
q_n recurrence (order 4, deg 19)	verified (74 terms)	§2
Descent membership of $s_n, \hat{P}_n$	conditional/compatible with anchors	§3
$s_3, \hat{P}_3$ exact values	conditional on membership	§3
P_3 exact value	conditional (2 hypotheses)	§4
Sector measurement (decay rates)	conditional/measured	§5
Phase-2 assembly from (DWORK $_{\mathbb{Z}}$)	proven conditionally; verified data	§6
(DWORK), incl. exceptional normalization	open (new)	§6

2 The exact recurrence and its two sectors

Finding 1 (verified candidate). The computed sequence q_n of the totally symmetric $M_{0,10}$ cell is annihilated, throughout the available exact range, by a linear recurrence of order 4 with polynomial coefficients of degree 19 (leading/trailing coefficients palindromic, $7381728 \cdot [1, -6340, 67974, -6340, 1]$), reconstructed from 105 terms by 3-prime CRT ($\approx 2^{125}$) with exact rational lifting, and certified to annihilate all 74 exact values of q_n (all 70 testable offsets vanish identically; independently re-verified). The characteristic polynomial

$$\chi(\lambda) = \lambda^4 - 6340\lambda^3 + 67974\lambda^2 - 6340\lambda + 1 = (\lambda^2 - a\lambda + 1)(\lambda^2 - b\lambda + 1), \quad a, b = 3170 \pm 1824\sqrt{3},$$

factors over $\mathbb{Q}(\sqrt{3})$ (check: $a + b = 6340$, $2 + ab = 67974$). The four roots form two reciprocal pairs (Poincaré-duality pairing): *sector A* $\{6329.2605\dots, 1.57996\dots \times 10^{-4}\}$ and *sector B* $\{10.6454\dots, 0.0939374\dots\}$.

The order matches the motivic rank ($\zeta(3)$: order 2; $\zeta(5)$: order 3; $\zeta(7)$: order 4), while the degree jump ($9 \rightarrow 19$ from weight 5) shows arithmetic complexity growing faster than rank. Annihilation of finitely many terms is not a proof that the operator governs q_n for all n ; a creative-telescoping certificate was attempted on two representations (an 8-variable low-coupling window model and the 4-fold hypergeometric-sum form of §4) and in both cases hit the geometric elimination wall (successive eliminations $\geq 3h$ each); checkpoints are preserved. The recurrence’s status therefore remains *verified*, with overwhelming structural support (motivic order match, palindromy, two independent window representations agreeing on all computed terms).

3 Two period ladders

Finding 2 (conditional, compatible with all available anchors). The weight-5 descent data $(q_n, s_n, \hat{P}_n)$ — equivalently the form I''_n — ride the order-4 operator: the three known values of each, augmented by the operator’s trailing-coefficient degeneracy at the initial offset (which also re-derives $q_3 = 94357501$ from q_0, q_1, q_2 , an internal consistency check), determine

$$s_3 = \frac{1396906795}{3}, \quad \hat{P}_3 = \frac{23217557999}{972} \quad (972 = 2^2 3^5).$$

The primitive data (P_n, I'_n) do *not* satisfy the operator: propagation is inconsistent (the naive P_3 candidate even acquires a spurious prime factor 107). Weight seven thus splits into two holonomic species — the descended and the primitive ladder — which weight five does not separate.

The values $s_3, \hat{P}_3$ are exact *conditional on descent membership* (that the actual $s_n, \hat{P}_n$ satisfy the operator for all n); membership is compatible with every available anchor and is suggested by the motivic rank structure, but a period-level telescoping certificate is the outstanding closure (see §7).

4 The value of P_3 by denominator snap

The full cellular integral admits the exact all-positive 4-fold series representation of [6] (verified against the anchors), hence $I_n > 0$ and $I_{n+1} < I_n$ rigorously. The observed values collapse: $I_0 = 3.5554\dots$, $I_1 = 3.2071\dots \times 10^{-5}$, $I_2 = 1.05313\dots \times 10^{-9}$ — the full form is a near-perfect cancellation $I' \approx -\zeta(2)I''$ (four orders at $n = 2$).

Proposition 3 (snap; conditional on (i)–(ii)). *Assume (i) $\text{den}(P_3) \mid 12d_3^7$ and (ii) descent membership (so $s_3, \hat{P}_3$ are as above). Then*

$$P_3 = \frac{23478462179525}{69984}, \quad \text{den}(P_3) = 69984 = 2^5 3^7.$$

Proof (no direct evaluation of I_3). By the filtration identity, $P_3 = [\frac{75}{4}q_3\zeta(7) - 3s_3\zeta(5) + \zeta(2)I_3''] - I_3$, where the bracket is computable to arbitrary precision from (ii) and $I_3'' = -9q_3\zeta(5) + 2s_3\zeta(3) - \hat{P}_3$. The grid $(1/12d_3^7)\mathbb{Z}$ of hypothesis (i) has spacing $h = 2.977 \times 10^{-7}$. Certified rational enclosures for the zeta values place the bracket strictly between the stated grid point and that grid point plus $h/2$. Independently, positivity together with the rigorous series majorant gives $0 < I_3 \leq 3.09 \times 10^{-9} < h/2$. Hence P_3 is the unique grid point in the permitted one-sided interval below the bracket, and it is the stated rational. The decimal proximity is not the certification: it only indicates which short certified enclosure to check. $\square$

Independent verification. The implied value $I_3 = 5.6299224184893 \times 10^{-14}$ was tested against the all-positive series: partial sums (rigorous lower bounds) climb to 99.75% of it without ever exceeding it, and power-law extrapolation lands within 0.04%; a proved separable majorant gives the rigorous bound $I_3 \leq 3.09 \times 10^{-9} < h/2$, making the half-window premise unconditional (the snap still assumes (i)–(ii)). Consistency locks: the residual is positive as required; $I_3/I_2 = 5.35 \times 10^{-5}$ continues the ratio sequence 9.0×10^{-6} , 3.28×10^{-5} monotonically toward the sector-A root 1.58×10^{-4} , as duality predicts for the full form; a wrong grid hypothesis would land within 5.6×10^{-14} of a grid point; under a uniform-phase heuristic this has probability $\sim 2 \times 10^{-7}$.

Corollary 4 (conditional). $\text{den}(P_3) = 2^5 3^7$ divides $d_3^7 = 2^7 3^7$: the naive inclusion $d_3^7 P_3 \in \mathbb{Z}$ holds at $n = 3$, with ord_3 tight and ord_2 slack 2. At its first 3-adic opportunity the weight-7 primitive ladder shows no 12-type correction — consistent with the $n \leq 2$ data and with $\text{den}(\hat{P}_3) = 2^2 3^5$ on the descent ladder: the anomalous correction factor measured at weight 5 [5] does not recur in this family through $n = 3$.

5 Sector measurement and the Diophantine verdict

For a weight-7 form with denominator growth $d_n^{7+o(1)}$, irrationality-type conclusions require decay below $e^{-7} = 9.119 \times 10^{-4}$. The two sub-unit roots straddle this threshold: sector A's 1.58×10^{-4} passes (margin $e^{1.753}$), sector B's 0.0939 fails by two orders.

Finding 5 (measured, conditional on the recurrence and descent membership). Propagating I_n'' through the exact operator at 300-digit precision from its four exact initial values gives ratios increasing monotonically 0.0042, 0.0181, 0.0306, $\dots$, 0.0833 ($n = 29$) toward 0.0939: the *descended* ladder rides sector B. The filtration then locks the primitive projection: since I_n (sector A, ratios $9.0 \times 10^{-6}, 3.3 \times 10^{-5}, 5.4 \times 10^{-5} \rightarrow 1.58 \times 10^{-4}$) is negligible against $\zeta(2)I_n''$, we get $I_n' \approx -\zeta(2)I_n''$, and indeed $I_3'/I_2' = 0.0306408$ against $I_3''/I_2'' = 0.0306408$ (difference 2.8×10^{-8}). *Both projections ride sector B.*

Corollary 6 (conditional on the recurrence, sector assignment, descent membership, and stated denominator cost). *The standard simultaneous-linear-form criterion applied to these two projections cannot prove “at least one of $\zeta(5), \zeta(7)$ is irrational” under a d_n^7 -type denominator cost: the required decay e^{-7} is missed by the factor $0.0939/e^{-7} \approx 103$. The $\zeta(2)$ -elimination costs a decay factor $0.0939/1.58 \times 10^{-4} \approx 594$ — an archimedean analogue of the Betti elimination cost of [5]. The surviving Diophantine route is asymmetric: optimize $\mathcal{M}(a) = -\log \rho_{\text{proj}}(a) - \kappa(a)$ over the parameter orbit (decay and denominator cost move together; positive $\mathcal{M}$ is the Diophantine region).*

6 Phase 2 of the central-binomial law: reduction to a Dwork congruence

We now return to weight 5. With Zudilin’s sequences $q_n, p_n, \tilde{p}_n$ [2] and $A_n = 2d_n^5 p_n \in \mathbb{Z}$, the corrected symmetric law $12d_n^5 P_n \in \mathbb{Z}$ is equivalent to $\binom{2n}{n} \mid 6A_n$, i.e. $\text{ord}_p(p_n) \geq \kappa - 5L$ at every prime $p \geq 5$, where $\kappa = \text{ord}_p \binom{2n}{n}$ (Kummer’s carry count) and $L = \text{ord}_p d_n = \lfloor \log_p n \rfloor$. Phase 1 ($n < p \leq 2n$, $L = 0$) is proven in [4]. For $p \leq n$ the following structure was discovered and verified in this campaign.

Finding 7 (verified; 272 descents, 46 band pairs, 0 failures). Let $a = \lfloor n/p \rfloor$. Then

$$p^5 \frac{p_n}{q_n} \equiv \frac{p_a}{q_a}, \quad p^3 \frac{\tilde{p}_n}{q_n} \equiv \frac{\tilde{p}_a}{q_a} \pmod{p^k}, \quad k \geq 2 - \kappa \text{ (typical } 3 - \kappa),$$

for all tested (n, p) , $p \geq 5$, $n \leq 45$. The digit constants are exact ladder values — in particular

$$\frac{p_1}{q_1} = \frac{29}{28}, \quad \frac{\tilde{p}_1}{q_1} = \frac{101}{84}, \quad \frac{p_2}{q_2} = \frac{24289}{23424},$$

identified first as the unique low-height fit to the measured p -adic digits and then confirmed exactly. The first correction digit is antisymmetric in $r = n - ap$ about $(p-1)/2$ — the derivative signature of a Dwork $F(x)/F(x^p)$ congruence. Plain Lucas congruences $q_{ap+b} \equiv q_a q_b \pmod{p}$ *fail* at every tested prime: the operative structure is a Frobenius descent of the ratio of solutions, not a digit product. At exceptional primes dividing a digit-constant denominator (e.g. $7 \mid 28$), the ratio statement degrades but the combined integer invariant $\text{ord}_p(p_n) \geq \kappa - 5L$ is preserved in every tested case.

For the all-prime assembly it is useful to distinguish the observed ratio congruence from the integer-lattice statement that it must induce. Put $\kappa_p(m) = \text{ord}_p \binom{2m}{m}$, and write $a = \lfloor n/p \rfloor$. The required denominator-free form is the one-step inequality

$$\text{ord}_p(p_n) - \kappa_p(n) \geq \text{ord}_p(p_a) - \kappa_p(a) - 5. \quad (\text{DWORK}_{\mathbb{Z}})$$

Along a clean descent path, (DWORK) together with the free divisibility of q_n already gives the final bound directly. The stronger one-step form (DWORK $_{\mathbb{Z}}$) also records transport of any excess

divisibility of q_a . At an exceptional prime it asserts, in particular, that the excess divisibility of q_n compensates for the denominator of the relevant digit ratio. That compensation is verified in the finite data but is part of the open theorem; mere cross-multiplication of the displayed ratio congruence does not by itself prove it.

Proposition 8 (assembly; proven modulo (DWORK $_{\mathbb{Z}}$)). *Assume (DWORK $_{\mathbb{Z}}$) for every $p \geq 5$ and every $n \geq p$. Then*

$$\text{ord}_p(p_n) \geq \kappa_p(n) - 5\lfloor \log_p n \rfloor \quad (p \geq 5).$$

Consequently Phase 2 of the central-binomial law holds. Combined with Phase 1 [4], this yields $\binom{2n}{n} \mid 6A_n$ and hence $12d_n^5 P_n \in \mathbb{Z}$ for all n .

Proof. Set $n_0 = n$ and $n_{j+1} = \lfloor n_j/p \rfloor$. For $L = \lfloor \log_p n \rfloor$ we have $n_L < p$. Applying (DWORK $_{\mathbb{Z}}$) successively gives

$$\text{ord}_p(p_n) - \kappa_p(n) \geq \text{ord}_p(p_{n_L}) - \kappa_p(n_L) - 5L.$$

Here $\kappa_p(n_L) = 0$, and p_{n_L} is p -integral because its known denominator divides a power of d_{n_L} , with $p > n_L$. The right-hand side is therefore at least $-5L$, proving the claim. $\square$

Both the one-step form (DWORK $_{\mathbb{Z}}$) and the resulting inequality $\text{ord}_p(p_n) \geq \kappa - 5L$ are verified with zero failures for all $p \geq 5$, $n \leq 45$ — including the 206 zero-slack tight cases of the earlier ledger, which the descent now explains: the d_n^5 reserve pays exactly 5 orders per digit level, and κ carries the Kummer carries.

Literature status and proof program. A targeted audit (with searches of the primary sources) indicates (DWORK) is *not* covered off-the-shelf: Delaygue–Rivoal–Roques [7] treat hypergeometric coefficient sequences and mirror-map quotients, not harmonic-companion ratios; Malik–Straub [8] prove scalar Lucas-type results. Derivative-corrected Lucas congruences for the classical Apéry sequence [9] are a close precedent for the observed antisymmetric correction, but still concern the leading sequence rather than a coefficientwise ratio of independent solutions. Matrix Dwork theory [10] supplies the correct higher-rank architecture, not an off-the-shelf theorem for p_n/q_n . The planned proof route: (1) restate (DWORK) denominator-free in the period vector $(q_a, p_a, \tilde{p}_a)$ and prove (DWORK $_{\mathbb{Z}}$), so exceptional primes become explicit lattice bookkeeping; (2) prove the $a = 1$ band by splitting the very-well-poised summand into p -blocks and expanding factorial and harmonic blocks through weight 5 (Wolstenholme-block cancellations); (3) treat $(p_n, \tilde{p}_n)$ simultaneously, exposing a triangular Frobenius matrix; (4) generalize in a . Phase 1 is the proven terminal boundary condition for the iteration. It should not be called the $a = 0$ case of the displayed ratio descent: because of the prefactor p^5 , that formal congruence alone is too weak to recover Phase 1. To place the weight-5 statement inside matrix Dwork theory one still needs a period-matrix or diagonal realization of the complete Zudilin triple. The period-level telescoping project for the distinct weight-7 family (§2) may provide a useful model for that step, but does not itself supply the required weight-5 realization.

7 Outlook

Three closures would upgrade the night’s conditional results to theorems: (1) a creative-telescoping certificate (or alternative proof) that the order-4 operator governs the weight-7 family at period level — closing descent membership, hence $s_3, \hat{P}_3$, hence hypothesis (ii) of the snap; (2) a proof of the denominator-grid hypothesis (i), e.g. by the Krattenthaler–Rivoal-type arithmetic for the weight-7 forms; (3) a proof of (DWORK) together with its integer-lattice form (DWORK $_{\mathbb{Z}}$), completing the

corrected Brown–Zudilin law $12d_n^5 P_n \in \mathbb{Z}$. Independent of all three, the sharpest new open object is the vector-valued Dwork descent itself: one congruence family, verified across 272 instances, whose archimedean limit is the classical convergence $p_n/q_n \rightarrow \zeta(5)$ and whose p -adic content is a self-similarity under base- p digit shift. One ratio, all completions, one mechanism.

Provenance

Results obtained 17–18 July 2026 in a collaborative session: coordination, reductions and proofs as cited; exact computation by fleet agents; independent review by a second model (in particular the conditional-rigor form of Proposition 3 and the literature audit of §6). Complete scripts and exact data in the repository (`worthiness/`): recurrence and terms (`zeta7_q_recurrence.json`, `zeta7_lc_terms.txt`), snap and verification (`ZETA7_P3_SNAP.md`, `ZETA7_P3_VERIFICATION.md`, `zeta7_p3_*.py`), sector measurement (`zeta7_sector_measurement.py`), Phase-2 program (`PHASE2_BAND_BLUEPRINT.md`, `PHASE2_BAND_V.md`, `lemma_cb_band_v*.py`).

References

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